Solution to Problem 1.

(a) i $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ where $\boldsymbol{U} \in \mathbb{R}^{m\times m}$ and $\boldsymbol{V} \in \mathbb{R}^{n\times n}$ are unitary, i.e., $\boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}$ and $\boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}$, and $\boldsymbol{\Sigma} \in \mathbb{R}^{m\times n}$ is diagonal. (The convention is that the diagonal elements of $\boldsymbol{\Sigma}$ are sorted from the largest to the smallest.) [3]

ii A. $\boldsymbol{U}_r \in \mathbb{R}^{m\times r}$, $\boldsymbol{\Sigma}_r \in \mathbb{R}^{r\times r}$, and $\boldsymbol{V}_r \in \mathbb{R}^{r\times n}$. [2]

B. $\boldsymbol{U}_r = \boldsymbol{U}_{:,1:r}$, $\boldsymbol{\Sigma}_r = \boldsymbol{\Sigma}_{1:r,1:r}$, and $\boldsymbol{V}_r = \boldsymbol{V}_{:,1:r}$. [2]

iii $\boldsymbol{A}\boldsymbol{A}^T = \boldsymbol{U}_r\boldsymbol{\Sigma}_r^2\boldsymbol{U}_r^T$, and $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{V}_r\boldsymbol{\Sigma}_r^2\boldsymbol{V}_r^T$. [2]

iv $\boldsymbol{A}^\dagger = \boldsymbol{V}_r\boldsymbol{\Sigma}_r^{-1}\boldsymbol{U}_r^T$. (As $\boldsymbol{\Sigma}$ is a diagonal matrix with nonzero diagonal elements, $\boldsymbol{\Sigma}_r^{-1}$ is well defined and easy to compute.) [2]

v A. It suffices to show that $\boldsymbol{A}\boldsymbol{x}_0 = \boldsymbol{y}$. $\boldsymbol{y} \in \text{range}(\boldsymbol{U}_r)$ implies that $\boldsymbol{y} = \boldsymbol{U}_r\boldsymbol{c}$ for some nonzero vector $\boldsymbol{c} \in \mathbb{R}^r$.
Now we compute $\boldsymbol{A}\boldsymbol{x}_0$:

$$\begin{aligned}\boldsymbol{A}\boldsymbol{x}_0 &= \boldsymbol{A}\boldsymbol{A}^\dagger\boldsymbol{y}\\ &= (\boldsymbol{U}_r\boldsymbol{\Sigma}_r\boldsymbol{V}_r^T)(\boldsymbol{V}_r\boldsymbol{\Sigma}_r^{-1}\boldsymbol{U}_r^T)(\boldsymbol{U}_r\boldsymbol{c})\\ &= \boldsymbol{U}_r\boldsymbol{\Sigma}_r\boldsymbol{I}_r\boldsymbol{\Sigma}_r^{-1}\boldsymbol{I}_r\boldsymbol{c}\\ &= \boldsymbol{U}_r\boldsymbol{c} = \boldsymbol{y}.\end{aligned}$$

Hence $\boldsymbol{x}_0 \in \mathcal{X}$. [2]

B. The subspace $\mathcal{X}_\perp$ has dimension $n - r$. One orthonormal basis of $\mathcal{X}_\perp$ is given by $\boldsymbol{V}_{:,r+1:n}$ where $\boldsymbol{V}$ is the right singular vector matrix of $\boldsymbol{A}$. [2]

C. For all $\boldsymbol{x} \in \mathcal{X}$, it holds that

$$\begin{aligned}\|\boldsymbol{x}\|_2^2 &= \langle \boldsymbol{x}, \boldsymbol{x}\rangle = \langle \boldsymbol{x}_0 + \boldsymbol{x}_\perp, \boldsymbol{x}_0 + \boldsymbol{x}_\perp\rangle\\ &= \|\boldsymbol{x}_0\|_2^2 + 2\langle \boldsymbol{x}_0, \boldsymbol{x}_\perp\rangle + \|\boldsymbol{x}_\perp\|_2^2\\ &\overset{(a)}{=} \|\boldsymbol{x}_0\|_2^2 + \|\boldsymbol{x}_\perp\|_2^2\\ &\geq \|\boldsymbol{x}_0\|_2^2,\end{aligned}$$

where (a) follows from the orthogonality between $\boldsymbol{x}_0$ and $\boldsymbol{x}_\perp$, and the last inequality becomes equality if and only if $\boldsymbol{x}_\perp = \boldsymbol{0}$. Hence $\|\boldsymbol{x}_0\|_2^2 = \min_{\boldsymbol{x}\in\mathcal{X}} \|\boldsymbol{x}\|_2^2$. [2]

(b) i It is straightforward to verify that

$$\boldsymbol{A}\boldsymbol{A}^T = \begin{bmatrix} 16 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

[2]

ii It is elementary to see that $\|\boldsymbol{v}\|_2 = 1$ and $\boldsymbol{A}\boldsymbol{v} = \boldsymbol{0}$. [2]

iii It can be verified that

$$\begin{aligned} \boldsymbol{A} &= \begin{bmatrix} 2 & 2 & 2 & 2 \\ 1 & 1 & -1 & -1 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\boldsymbol{U}} \underbrace{\begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\boldsymbol{\Sigma}} \underbrace{\frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}}_{\boldsymbol{V}^T} \end{aligned}$$

[2]

iv

$$\begin{aligned} \boldsymbol{A}^\dagger &= \boldsymbol{V}_r \boldsymbol{\Sigma}_r^{-1} \boldsymbol{U}_r \\ &= \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

[3]

v

$$\boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{y} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$$
$$= \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 3 \\ 1 \\ -1 \\ 1 \end{bmatrix}.$$

[2]

vi By inspection, it is clear that $\boldsymbol{x}' = [2, 0, 0, 0]^T$. [2]

Solution to Problem 2.

(a) i The complete OMP algorithm is stated in Algorithm 2.3.

Algorithm 2.3 OMP Algorithm

1: **Input**: S, $\boldsymbol{A}$, and $\boldsymbol{y}$.
2: **Initialisation**: $\boldsymbol{x} = \boldsymbol{0}$, $\mathcal{T}^0 = \phi$, and $\boldsymbol{y}_r = \boldsymbol{y}$.
3: **for** $l = 1, 2, \cdots, S$ **do**
4: $\quad i_l = \arg\max_i |\langle \boldsymbol{a}_i, \boldsymbol{y}_r \rangle|$ ▷ Find a new index
5: $\quad \mathcal{T}^l = \mathcal{T}^{l-1} \bigcup \{i_l\}$ ▷ Expand the support set $\mathcal{T}$
6: $\quad \boldsymbol{x}_{\mathcal{T}^l} = \boldsymbol{A}_{\mathcal{T}^l}^{\dagger} \boldsymbol{y}$ ▷ Update $\boldsymbol{x}$
7: $\quad \boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}$ ▷ Update $\boldsymbol{y}_r$

[2]

ii It is clear that

$$\begin{aligned} \boldsymbol{y}_r &= \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}^{l-1} \\ &= \boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^{l-1}} \boldsymbol{x}_{\mathcal{T}^{l-1}}^{l-1} \\ &= \boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^{l-1}} \boldsymbol{A}_{\mathcal{T}^{l-1}}^{\dagger} \boldsymbol{y}. \end{aligned}$$

Hence,

$$\begin{aligned} \boldsymbol{A}_{\mathcal{T}^{l-1}}^T \boldsymbol{y}_r &= \boldsymbol{A}_{\mathcal{T}^{l-1}}^T (\boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^{l-1}} \boldsymbol{A}_{\mathcal{T}^{l-1}}^{\dagger} \boldsymbol{y}) \\ &= \boldsymbol{A}_{\mathcal{T}^{l-1}}^T \boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^{l-1}}^T \boldsymbol{A}_{\mathcal{T}^{l-1}} \left((\boldsymbol{A}_{\mathcal{T}^{l-1}}^T \boldsymbol{A}_{\mathcal{T}^{l-1}})^{-1} \boldsymbol{A}_{\mathcal{T}^{l-1}}^T \right) \boldsymbol{y} \\ &= \boldsymbol{A}_{\mathcal{T}^{l-1}}^T \boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^{l-1}}^T \boldsymbol{y} = \boldsymbol{0}. \end{aligned}$$

Therefore, $i_l = \arg\max_i |\langle \boldsymbol{a}_i, \boldsymbol{y}_r \rangle| \notin \mathcal{T}^{l-1}$. [4]

iii A. The mutual coherence of $\boldsymbol{A}$ is defined as

$$\mu(\boldsymbol{A}) = \max_{i \neq j} |\langle \boldsymbol{a}_i, \boldsymbol{a}_j \rangle|.$$

[1]

B.

$$\begin{aligned} |\langle \boldsymbol{a}_i, \boldsymbol{y} \rangle| = \left| \boldsymbol{a}_i^T \left(\sum_{j \in \mathcal{T}_0} \boldsymbol{a}_j \boldsymbol{x}_{0,j} \right) \right| &\overset{(a)}{\leq} \sum_{j \in \mathcal{T}_0} \left| \boldsymbol{a}_i^T \boldsymbol{a}_j \right| |\boldsymbol{x}_{0,j}| \\ &\overset{(b)}{\leq} \mu \sum_{j \in \mathcal{T}_0} |\boldsymbol{x}_{0,j}| \overset{(c)}{\leq} \mu \sqrt{S} \, \|\boldsymbol{x}_0\|_2 , \end{aligned}$$

where (a) comes from triangle inequality, (b) follows from the definition of mutual coherence, and (c) holds due to Cauchy-Schwartz inequality. [3]

(b) i The complete SP algorithm is stated in Algorithm 2.4.

Algorithm 2.4 SP Algorithm

Input: S, $\boldsymbol{A}$, and $\boldsymbol{y}$.
Initialisation: $\mathcal{T}^0 = \text{supp}(H_S(\boldsymbol{A}^T\boldsymbol{y}))$, and $\boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^0}\boldsymbol{A}_{\mathcal{T}^0}^{\dagger}\boldsymbol{y}$.
while $l = 1, 2, \cdots$ until stop criteria hold **do**
  $\tilde{\mathcal{T}}^l = \mathcal{T}^{l-1} \bigcup \text{supp}(H_S(\boldsymbol{A}^T\boldsymbol{y}_r))$ ▷ Expand support
  Set $\boldsymbol{b}$ by $\boldsymbol{b}_{\tilde{\mathcal{T}}^l} = \boldsymbol{A}_{\tilde{\mathcal{T}}^l}^{\dagger}\boldsymbol{y}$ and $\boldsymbol{b}_{\left(\tilde{\mathcal{T}}^l\right)^c} = \boldsymbol{0}$ ▷ Estimate a $2S$-sparse signal
  Set $\mathcal{T}^l = \text{supp}(H_S(\boldsymbol{b}))$ ▷ Shrink support
  Set $\boldsymbol{x}$ by $\boldsymbol{x}_{\mathcal{T}^l} = \boldsymbol{A}_{\mathcal{T}^l}^{\dagger}\boldsymbol{y}$ and $\boldsymbol{x}_{\left(\mathcal{T}^l\right)^c} = \boldsymbol{0}$ ▷ Estimate an S-sparse signal
  $\boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}$ ▷ Update $\boldsymbol{y}_r$

[2]

ii
$$H_3([-9, 8, -7, 6, -5]^T) = [-9, 8, -7, 0, 0].$$

[2]

iii A matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ is said to satisfy the RIP with parameters (K, δ), if for all $\mathcal{T} \subset [n]$ such that $|\mathcal{T}| \leq K$ and for all $\boldsymbol{q} \in \mathbb{R}^{|\mathcal{T}|}$, it holds that

$$(1-\delta)\left\|\boldsymbol{q}\right\|_2^2 \leq \left\|\boldsymbol{A}_{\mathcal{T}}\boldsymbol{q}\right\|_2^2 \leq (1+\delta)\left\|\boldsymbol{q}\right\|_2^2.$$

The RIC δ_K is defined as the smallest constant δ for which the K-RIP holds, i.e.,

$$\delta_K = \inf\left\{\delta :\ (1-\delta)\left\|\boldsymbol{q}\right\|_2^2 \leq \left\|\boldsymbol{A}_{\mathcal{T}}\boldsymbol{q}\right\|_2^2 \leq (1+\delta)\left\|\boldsymbol{q}\right\|_2^2. \right.$$
$$\left. \forall\, |\mathcal{T}| \leq K,\ \forall \boldsymbol{q} \in \mathbb{R}^{|\mathcal{T}|}\right\}.$$

[4]

iv $-\delta_{2S} \leq \sigma\left(\boldsymbol{A}_{\mathcal{I}}^T\boldsymbol{A}_{\mathcal{J}}\right) \leq \delta_{2S}$ where $\sigma\left(\boldsymbol{A}_{\mathcal{I}}^T\boldsymbol{A}_{\mathcal{J}}\right)$ denotes the singular value of the matrix $\boldsymbol{A}_{\mathcal{I}}^T\boldsymbol{A}_{\mathcal{J}}$. [2]

Solution to Problem 3.

(a) i As the constraints in (3.1) involves m many equations, it is clear that $\boldsymbol{v} \in \mathbb{R}^m$. The Lagrangian is given by

$$L(\boldsymbol{x}, \boldsymbol{v}) = \frac{1}{2} \|\boldsymbol{x}\|_2^2 + \boldsymbol{v}^T(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}).$$

[2]

ii The dual function is given by

$$g(\boldsymbol{v}) = \min_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{v}) = \min_{\boldsymbol{x}} \frac{1}{2} \|\boldsymbol{x}\|_2^2 + \boldsymbol{v}^T(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}).$$

To find the closed form of $g(\boldsymbol{v})$, set $\frac{d}{d\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{v})$ to $\mathbf{0}$. Note that

$$\frac{d}{d\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{x} - \boldsymbol{A}^T\boldsymbol{v}.$$

It is clear that the optimal

$$\boldsymbol{x}^\star = \boldsymbol{A}^T\boldsymbol{v}.$$

Hence

$$\begin{aligned} g(\boldsymbol{v}) &= \frac{1}{2} \left\|\boldsymbol{A}^T\boldsymbol{v}\right\|_2^2 + \boldsymbol{v}^T(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{A}^T\boldsymbol{v}) \\ &= \boldsymbol{v}^T\boldsymbol{y} - \frac{1}{2}\boldsymbol{v}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{v}. \end{aligned}$$

[2]

iii The dual problem is given by

$$\max_{\boldsymbol{v}} g(\boldsymbol{v}) = \max_{\boldsymbol{v}} \boldsymbol{v}^T\boldsymbol{y} - \frac{1}{2}\boldsymbol{v}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{v}.$$

By setting $\frac{d}{d\boldsymbol{v}}\ g(\boldsymbol{v})$ to $\mathbf{0}$, one obtains

$$-\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{v}^\star + \boldsymbol{y} = \mathbf{0}.$$

Or equivalently

$$\boldsymbol{v}^\star = (\boldsymbol{A}\boldsymbol{A}^T)^{-1}\boldsymbol{y}.$$

[2]

iv From the above steps, it is clear that

$$\boldsymbol{x}^\star = \boldsymbol{A}^T\boldsymbol{v}^\star = \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{A}^T)^{-1}\boldsymbol{y}\ (= \boldsymbol{A}^\dagger\boldsymbol{y}).$$

[2]

(b) i It is clear that

$$\partial_x f(x) = x - y + \lambda \cdot \operatorname{sign}(x),$$

where

$$\operatorname{sign}(x) = \begin{cases} 1 & \text{if } x > 0, \\ [-1, 1] & \text{if } x = 0, \\ -1 & \text{if } x < 0. \end{cases}$$

Hence, it is straightforward to verify that

$$x^\star = \eta_\lambda(y) = \begin{cases} y - \lambda & \text{if } y > \lambda, \\ 0 & \text{if } y \in [-\lambda, \lambda], \\ y + \lambda & \text{if } y < -\lambda. \end{cases}$$

[4]

ii It is clear that

$$\nabla_{\boldsymbol{x}} g(\boldsymbol{x})|_{\boldsymbol{x}^k} = \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{y}).$$

It holds that

$$\begin{aligned} &\frac{1}{2}\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2 + \lambda\|\boldsymbol{x}\|_1 \\ &\approx \frac{1}{2}\left\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}^k\right\|_2^2 + \left\langle \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{y}), \boldsymbol{x} - \boldsymbol{x}^k \right\rangle + \frac{1}{2t_k}\left\|\boldsymbol{x} - \boldsymbol{x}^k\right\|_2^2 + \lambda\|\boldsymbol{x}\|_1 \\ &= c + \frac{1}{2t_k}\left\|\boldsymbol{x} - \boldsymbol{x}^k + t_k\boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{y})\right\|_2^2 + \lambda\|\boldsymbol{x}\|_1. \end{aligned}$$

The optimal solution of

$$\min_{\boldsymbol{x}} \frac{1}{2t_k}\left\|\boldsymbol{x} - \boldsymbol{x}^k + t_k\boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{y})\right\|_2^2 + \lambda\|\boldsymbol{x}\|_1$$

admits the closed form

$$\boldsymbol{x}^\star = \eta_{\lambda t_k}(\boldsymbol{x}^k - t_k\boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{y})).$$

Hence, the IST algorithm is given by the following iteration:

$$\boldsymbol{x}^{k+1} = \eta_{\lambda t_k}(\boldsymbol{x}^k - t_k \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{y})).$$

[3]

(c) i It is clear that (3.6) can be written as

$$\min_{\boldsymbol{x},\boldsymbol{z}} \|\boldsymbol{z}\|_1 \text{ s.t. } \begin{bmatrix} \boldsymbol{D} \\ \boldsymbol{I} \end{bmatrix} \boldsymbol{x} + \begin{bmatrix} \boldsymbol{0} \\ -\boldsymbol{I} \end{bmatrix} \boldsymbol{z} = \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{0} \end{bmatrix}.$$

Hence, $f(\boldsymbol{x}) = 0$, $g(\boldsymbol{z}) = \|\boldsymbol{z}\|_1$,

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{D} \\ \boldsymbol{I} \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} \boldsymbol{0} \\ -\boldsymbol{I} \end{bmatrix}, \text{ and } \boldsymbol{c} = \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{0} \end{bmatrix}.$$

[3]

ii

$$\begin{aligned} L_\rho(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{v}) = &\|\boldsymbol{z}\|_1 + \boldsymbol{v}^T(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}) \\ &+ \frac{\rho}{2}\|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}\|_2^2. \end{aligned}$$

[3]

iii It is straightforward to verify that

$$L_\rho(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{v}) = c + \|\boldsymbol{z}\|_1 + \frac{\rho}{2}\left\|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c} + \frac{1}{\rho}\boldsymbol{v}\right\|_2^2.$$

Then,

$$\begin{aligned} \boldsymbol{x}^{k+1} &= \arg\min_{\boldsymbol{x}} L_\rho(\boldsymbol{x}, \boldsymbol{z}^k, \boldsymbol{v}^k) \\ &= \boldsymbol{A}^\dagger\left(-\boldsymbol{B}\boldsymbol{z}^k + \boldsymbol{c} - \frac{1}{\rho}\boldsymbol{v}^k\right), \end{aligned}$$

and

$$
\begin{aligned}
\boldsymbol{z}^{k+1} &= \arg\min_{\boldsymbol{z}} L_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{z}, \boldsymbol{v}^k) \\
&= \arg\min_{\boldsymbol{z}} \ \|\boldsymbol{z}\|_1 + \frac{\rho}{2}\left\|-\boldsymbol{z} + \frac{1}{\rho}\boldsymbol{v}^k_{m+1:m+n}\right\|_2^2 \qquad (3.9) \\
&= \arg\min_{\boldsymbol{z}} \ \frac{1}{\rho}\|\boldsymbol{z}\|_1 + \frac{1}{2}\left\|\boldsymbol{z} - \frac{1}{\rho}\boldsymbol{v}^k_{m+1:m+n}\right\|_2^2 \\
&= \eta_{\frac{1}{\rho}}\left(\frac{1}{\rho}\boldsymbol{v}^k_{m+1:m+n}\right).
\end{aligned}
$$

[4]

Remark. Equation (3.9) is the key step to find the closed form solution for updating $\boldsymbol{z}$. It uses the special structure of the matrix $\boldsymbol{B}$.